

Saifeddine Mosrati

R&D Data Engineer Apprentice | AI & Machine Learning | Azure Cloud | Cybersecurity

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SUMMARY

Engineering student at EPITA (Cybersecurity & Systems track) and R&D Data Engineer apprentice at Mantu for the past year, working mostly on AI forecasting models. I build Machine Learning pipelines and IoT (Internet of Things) platforms on Microsoft Azure, with hands-on experience in penetration testing and API security. Goal: specialize in the security of artificial intelligence systems.

PROFESSIONAL EXPERIENCE

R&D Data Engineer (Apprenticeship) — Mantu Lab, Paris 09/2025 – Present

- Designed a Machine Learning pipeline forecasting a building's energy use (Smart Building R&D project): 89 signals from 6 types of IoT sensors, 3 horizons (1 hour, 1 day, 1 week).
- Benchmarked 12 models per horizon; selected XGBoost and CatBoost, tuned with Optuna and explained with SHAP; forecasting API (Flask) deployed on Azure Container Apps.
- Rebuilt 41 days of lost sensor data (historical averages, every rebuilt value flagged); developed the IoT management platform (Spring Boot, Thymeleaf) on Azure.

Web Security Pentester (Internship) — Futurnet 04/2025 – 09/2025

- Secured REST APIs (strong authentication, input validation, encryption) and ran internal penetration tests with audit reports.
- Identified and remediated OWASP Top 10 and recent CVE vulnerabilities; web hardening (CSP, HTTP headers, WAF).
- Contributed to ISO/IEC 27001 compliance of web services.

Full Stack Developer (Internship) — Futurnet 06/2024 – 10/2024

- Built a web platform (React, Node.js API, SQL and NoSQL databases) with continuous integration and cloud deployment.

Full Stack Developer & Data Analyst (Freelance) — Occasional assignments 2023 – 2025

- Web applications (React, Node.js, PostgreSQL, MongoDB) and data dashboards (Python, Pandas, Power BI).

TECHNICAL PROJECTS

- Secure Linux Kernel Module (C, EPITA)** — kernel-level networking, hashing and symmetric encryption, system-call interception.
- AI Dockerfile Optimizer (AI & DevOps)** — analyzes a Dockerfile and generates an optimized, hardened version with an AI model.
- CyberPulse (cybersecurity)** — real-time dashboard visualizing cyberattacks by protocol, origin and type.
- Route Optimization (Python, graph theory, EPITA)** — multi-vehicle Chinese Postman on real Montreal data, min-cost flow.

EDUCATION

EPITA, Paris — MEng, Cybersecurity & Systems track (apprenticeship) 2025 – 2028

ESILV, La Défense — BSc Digital Engineering & Cybersecurity, **top of class**; Erasmus EPHEC 2022 – 2025

TECHNICAL SKILLS

AI & Data: Python, Machine Learning, Pandas, time series, Power BI

Cloud & DevSecOps: Microsoft Azure, Azure DevOps, Docker, CI/CD, Git, REST APIs

Cybersecurity: web penetration testing, API security, OWASP Top 10, vulnerability analysis (CVE), ISO/IEC 27001

Languages & Frameworks: C, C++, Java, Python, Bash, SQL, TypeScript, Spring Boot, React, Node.js

Systems: Linux (Debian, Kali), Raspberry Pi, Arduino

AWARDS, CERTIFICATIONS & LANGUAGES

Winner, Hussar Academy Hackathon (Paris 2024, Préfecture de Police)

Certification: Cisco – Junior Cybersecurity Analyst Career Path

Languages: Arabic (native), French (fluent), English (C1, TOEIC), Italian (B1)